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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/764,337	07/04/2024	Yueqiang WANG	20608-0055US01	8323

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EXAMINER
SAHLE, MAHIDERE S

ART UNIT	PAPER NUMBER
2872	

NOTIFICATION DATE	DELIVERY MODE
07/29/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Non Final-OA
2 mon from 3-mon due date: 2026-08-29
1 mon from 3-mon due date: 2026-09-29
3-mon due date: 2026-10-29
Due date w/1-mon EOT: 2026-11-29
Due date w/2-mon EOT: 2026-12-29
Final due date w/3-mon EOT (non-extendable): 2027-01-29
Check IDS when filing response
Input by: Qin ZHANG
Reviewed by: Qiwen MA

Office Action Summary	Application No. 18/764,337	Applicant(s) WANG et al.	
	Examiner MAHIDERE S SAHLE	Art Unit 2872	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04 July 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 04 July 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 09/24/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

Acknowledgment is made of receipt of Information Disclosure Statement (PTO-1449) filed 09/24/2024. An initialed copy is attached to this Office Action.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claims **1-15** are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Liu et al. (USPG Pub No. 2020/0344542 *as translation for WO/2019/134377*).

The applied reference has a common inventor with the instant application. Based upon the earlier effectively filed date of the reference, it constitutes prior art under 35 U.S.C. 102(a)(2). This rejection under 35 U.S.C. 102(a)(2) might be overcome by: (1) a showing under

37 CFR 1.130(a) that the subject matter disclosed in the reference was obtained directly or indirectly from the inventor or a joint inventor of this application and is thus not prior art in accordance with 35 U.S.C. 102(b)(2)(A); (2) a showing under 37 CFR 1.130(b) of a prior public disclosure under 35 U.S.C. 102(b)(2)(B) if the same invention is not being claimed; or (3) a statement pursuant to 35 U.S.C. 102(b)(2)(C) establishing that, not later than the effective filing date of the claimed invention, the subject matter disclosed in the reference and the claimed invention were either owned by the same person or subject to an obligation of assignment to the same person or subject to a joint research agreement.

Regarding claim 1, Liu discloses a glasses (see Fig. 4), comprising: a glasses frame (30) including a glasses rim (31) and two glasses temples (32), the two glasses temples (32) being rotatably connected to the glasses rim (see Fig. 4), respectively; and two speakers (20), wherein for each of the two speakers, the speaker includes an earphone core and a core housing (see Figs. 10-12, Paragraphs 4, 68 – it is inherent for the speaker to have an earphone core within a housing in order to produce sound/vibration), the speaker (20) is connected to one of the two glasses temples (32) via a hinge component (40) of the glasses temple connected thereto (see Fig. 16), the hinge component (40) is rotatable to change a position of the speaker relative to the glasses temple (32) (see Fig. 16), wherein the hinge component includes a hinge (40), a rod-shaped member (321), and a fixing member (323) (see Fig. 25, Paragraph 146), wherein: the rod-shaped member (321) is provided with a hinge chamber (3211) (see Fig. 25, Paragraph 146), the rod-shaped member (321) includes a first insertion hole (3212) at a side wall of the rod-shaped member (321) to communicate with the hinge chamber (3211) (see Fig. 25, Paragraph 146), the hinge (40) is at least partially inserted into the hinge chamber (3211) (see Fig. 24, Paragraph 142), and the hinge (40) includes a second insertion hole (421) corresponding to the first insertion hole (3212) (see Fig. 25, Paragraph 147), and the hinge (40) is connected to the rod-shaped member (321) through the fixing member (323) inserted into the first insertion hole (3212) and the second insertion hole (421) (see Fig. 25, Paragraphs 147,

148), and the one of the two glasses temples (32) includes at least one of a control circuit or a battery (Paragraph 156), the at least one of the control circuit or the battery being configured to drive the earphone core of the speaker (20) to vibrate to generate sound (Paragraphs 55, 156).

Regarding claim 2, Liu discloses the hinge (40) includes: a hinge mount (42) (Paragraph 140); a hinge arm (43) rotatably connected to the hinge mount (42) via a rotating shaft (41) (Paragraph 140), and being rotatable relative to the hinge mount (42) when an external force is applied to the hinge arm (43) to change the position of the speaker (20) relative to the glasses temple (32) (Paragraphs 82, 145); a support member (44) flexibly disposed on the hinge mount (42) (see Fig. 34, Paragraph 177); and an elastic member (45) configured to elastically offset the support member (44) toward the hinge arm (43) so that the support member elastically abuts on the hinge arm (see Fig. 34, Paragraphs 177, 178).

Regarding claim 3, Liu discloses wherein the second insertion hole (421) penetrates both sides of the hinge mount (42) (Paragraph 149).

Regarding claim 4, Liu discloses wherein the hinge arm (43) includes a first support surface (432) and a second support surface (433) connected to each other (see Fig. 33, Paragraph 177), the support member (44) includes a third support surface (441) (see Fig. 33, Paragraph 177), the third support surface (441) elastically abuts on the first support surface (432) or the second support surface (433) and when the hinge arm (43) rotates relative to the hinge mount (42) due to the external force (Paragraph 178), a connection between the first support surface (432) and the second support surface (433) drives the support member (44) against the elastic offset of the elastic member (45) to move in an opposite direction so that the third support surface (441) is switched from being elastically abutting on one of the first support surface (432) and the second support surface (433) to being elastically abutting on the other of the first support surface and the second support surface (see Fig. 33, Paragraph 178).

Regarding claim 5, Liu discloses wherein in a section perpendicular to a central axis of the rotating shaft (41), a ratio of a maximum distance between the rotating shaft (41) and the

connection to a shortest distance between the rotating shaft (41) and the first support surface (432) is in a range from 1.1 to 1.5 (Paragraphs 189, 190).

Regarding claim 6, Liu discloses wherein an included angle between the hinge mount (42) and the hinge arm (43) decreases when the third support surface (441) is switched from elastically abutting on the first support surface (432) to elastically abutting on the second support surface (433) (Paragraph 196).

Regarding claim 7, Liu discloses wherein an external force required when the third support surface (441) is switched from elastically abutting on the first support surface (432) to elastically abutting on the second support surface (433) is different from an external force required when the third support surface (441) is switched from elastically abutting on the second support surface (433) to elastically abutting on the first support surface (432) (Paragraph 210).

Regarding claim 8, Liu discloses wherein in a section perpendicular to a central axis of the rotating shaft (41), the connection has an arc shape (Paragraph 205).

Regarding claim 9, Liu discloses wherein the connection has a circular arc shape, and a curvature of the circular arc is between 5 and 30 (Paragraph 205).

Regarding claim 10, Liu discloses wherein in a section perpendicular to a central axis of the rotating shaft (41), an included angle between the first support surface (432) and the second support surface (433) is an obtuse angle (Paragraph 200).

Regarding claim 11, Liu discloses wherein the hinge mount (42) includes a mount body (423), a first lug (424), and a second lug (425), the first lug and the second lug being protruding from the mount body and spaced apart from each other (see Fig. 37, Paragraph 217), the hinge arm (43) includes an arm body (435) and a third lug (436) protruding from the arm body (see Fig. 37, Paragraph 217), and the third lug (436) is inserted into an interval region between the first lug (424) and the second lug (425), and rotatably connected to the first lug and the second lug via the rotating shaft (41) (see Fig. 37, Paragraph 217).

Regarding claim 12, Liu discloses wherein at least part of the support member (44) is disposed inside the interval region and located at a side of the third lug (436) facing the mount body (423) (see Fig. 37, Paragraph 217), the mount body (423) includes an accommodation chamber (4231) communicating with the interval region (see Fig. 37, Paragraph 217), the elastic member (45) is disposed inside the accommodation chamber (4231) (see Fig. 37, Paragraph 217), and the support member (44) elastically offsets toward the third lug (436) (see Fig. 37, Paragraph 217).

Regarding claim 13, Liu discloses wherein the core housing includes a contact surface, the contact surface includes a gradient structure, the gradient structure includes at least one convex portion or at least one concave portion (see Figs. 10-12, Paragraphs 67-69).

Regarding claim 14, Liu discloses wherein the gradient structure is located at a center or an edge of the contact surface (see Figs. 10-12, Paragraphs 67-69).

Regarding claim 15, Liu discloses wherein the contact surface is in direct or indirect contact with the user (see Figs. 10-12, Paragraphs 67-69), and the contact surface includes a first contact surface region and a second contact surface region (see Figs. 10-12, Paragraphs 67-69), a protrusion degree of the second contact surface region being greater than a protrusion degree of the first contact surface region (see Figs. 10-12, Paragraphs 67-69).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims **16-20** are rejected under 35 U.S.C. 103 as being obvious over Liu (USPG Pub No. 2020/0344542) in view of Chen et al. (CN 106954150 A), hereinafter “Chen”.

The applied reference has a common inventor with the instant application. Based upon the earlier effectively filed date of the reference, it constitutes prior art under 35 U.S.C. 102(a)(2).

This rejection under 35 U.S.C. 103 might be overcome by: (1) a showing under 37 CFR 1.130(a) that the subject matter disclosed in the reference was obtained directly or indirectly from the inventor or a joint inventor of this application and is thus not prior art in accordance with 35 U.S.C.102(b)(2)(A); (2) a showing under 37 CFR 1.130(b) of a prior public disclosure under 35 U.S.C. 102(b)(2)(B); or (3) a statement pursuant to 35 U.S.C. 102(b)(2)(C) establishing that, not later than the effective filing date of the claimed invention, the subject matter disclosed and the claimed invention were either owned by the same person or subject to an obligation of assignment to the same person or subject to a joint research agreement. See generally MPEP § 717.02.

Regarding claim 16, Liu discloses the claimed invention, but does not specify wherein the earphone core includes a transducing device disposed inside an accommodation chamber with an opening formed by the core housing, and being capable of generating a vibration based on an audio signal; a vibration transmission plate connected to the transducing device and exposed from the opening to transmit the vibration; and a vibration transmission layer covering an outer surface of the vibration transmission plate in an integral injection molding manner, and further being connected to the core housing to seal the opening. In the same field of endeavor, Chen discloses wherein the earphone care includes a transducing device (1802) disposed inside an accommodation chamber with an opening formed by the core housing, and being capable of generating a vibration based on an audio signal; a vibration transmission plate (1801) connected to the transducing device and exposed from the opening to transmit the vibration; and a vibration transmission layer (1830) covering an outer surface of the vibration transmission plate in an integral injection molding manner, and further being connected to the core housing to seal the opening (see Figs. 2-C, 6, 9, 18-A, 18-B, 20, 22-A, 23, 25, 30). Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to provide the glasses of Liu with wherein the earphone core includes a transducing device disposed inside an accommodation chamber with an opening formed by the core housing, and being capable of generating a vibration based on an audio signal; a vibration transmission plate connected to the transducing device and exposed from the opening to transmit the vibration; and a vibration transmission layer covering an outer surface of the vibration transmission plate in an integral injection molding manner, and further being connected to the core housing to seal the opening of Chen for the purpose of improving the tone quality of the speakers (Abstract).

Regarding claim 17, Liu and Chen teach the glasses set forth above for claim 16, Chen further discloses wherein the vibration transmission plate (1801) protrudes from the opening, and the vibration transmission layer (1830) is fitted with the outer surface of the vibration

transmission plate (1801) in an integral injection molding manner (see Figs. 2-C, 6, 9, 18-A, 18-B, 20, 22-A, 23, 25, 30). It would have been obvious to one of ordinary skill to provide the glasses of Liu with the teachings of Chen for at least the same reasons as those set forth above with respect to claim 16.

Regarding claim 18, Liu and Chen teach the glasses set forth above for claim 16, Chen further discloses wherein the vibration transmission plate (1801) is spaced apart from the core housing at the opening to form an annular spaced region around the vibration transmission plate, and the vibration transmission layer (1830) includes at least one through hole at a region corresponding to the annular spaced region (see Figs. 2-C, 6, 9, 18-A, 18-B, 20, 22-A, 23, 25, 30). It would have been obvious to one of ordinary skill to provide the glasses of Liu with the teachings of Chen for at least the same reasons as those set forth above with respect to claim 16.

Regarding claim 19, Liu and Chen teach the glasses set forth above for claim 18, Chen further discloses wherein the at least one through hole includes a plurality of through holes, the plurality of through holes being arranged around the vibration transmission plate at intervals (see Figs. 2-C, 6, 9, 18-A, 18-B, 20, 22-A, 23, 25, 30, Pgs. 27, 56). It would have been obvious to one of ordinary skill to provide the glasses of Liu with the teachings of Chen for at least the same reasons as those set forth above with respect to claim 16.

Regarding claim 20, Liu and Chen teach the glasses set forth above for claim 16, Chen further discloses wherein the vibration transmission plate includes hard rubber, and the vibration transmission layer includes soft rubber (Pgs. 10, 19). It would have been obvious to one of ordinary skill to provide the glasses of Liu with the teachings of Chen for at least the same reasons as those set forth above with respect to claim 16.

Prior Art Citations

Gasparetto (USPG Pub No. 2013/0201438 – Figs. 3a-3c) is being cited herein to show glasses relevant to the claimed invention.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MAHIDERE S SAHLE whose telephone number is (571)270-3329. The examiner can normally be reached Monday-Thursday 8:00 AM to 5:00 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ricky Mack can be reached at 571 272-2333. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/MAHIDERE S SAHLE/
Primary Examiner, Art Unit 2872
7/25/2026